Where the uplift jumps and where it lands: the BPS uplift anywhere in the fortuity window

Progress report, 26 October 2026

`syk_gravity` project

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Abstract

The report of 13 October solved the BPS uplift of one-flavor $\mathcal{N} = 2$ SYK from N to $N' = N + 1$ modes exactly. It did so at half filling, where one fact simplifies everything: the pair operator $E = \eta\wedge$ that couples the new mode maps old BPS states out of the fortuity window, so its images are exact. Here we remove that restriction and treat every sector (N', ℓ') of the window. The dimensional bookkeeping uses the per-degree Hodge tables of Kanazawa–Wettig (KW), which we review and check against our code.

1. *A jump map.* Each uplift starts from two slots of the old model, at levels $\ell' \pm 1$. Its first-order dressing visits level $\ell' + 2q - 1$ (or $\ell' - 2q + 1$) before returning. Where that intermediate level sits relative to the window sorts all sectors into three regimes, which we draw on CV's BPS-dimension plane (Figs. 1, 2): *central* ($|\ell'| \le q - 2$: gapped dressing, the regime of the 13 October report); *resonant* ($|\ell'| = q - 1$: the dressing passes through the gapless window-edge multiplet, the one KW single out as GOE/GSE); *edge* ($|\ell'| = q$: the dressing lands on BPS states, and CV's connecting map δ obstructs).
2. *The exact uplift holds everywhere,* restricted to the admissible sources $\ker\delta$. Every closed form of 13 October carries over unchanged: orthogonal splitting, minimum-norm uplift, closed-form decoder, monotone self-dressing. So does a particle–hole mirror symmetry $\lambda \mapsto 1 - \lambda$ between ℓ' and $-\ell'$. We verify all of this to machine precision against direct computations, for $q = 3$ and $q = 5$ at $N' \le 14$, and through the dimension count and the 13 October values at $N' = 15, 16$.
3. *Exact ranks of δ, and exceptional states as a radical.* A duality pairs the connecting maps at levels ℓ and $-\ell - 2(q - 1)$. It fixes the rank of δ in every sector. One step off the centre there is an *exchange*: the old exceptional states replace an equal number of obstructed ordinary classes. At the edge, the new exceptional states are the radical of the bilinear form $\int \beta \wedge \eta \wedge \alpha$, which is antisymmetric exactly when $(N' - q)/2$ is odd. This reproduces KW's Kramers counting. For $q = 5$ it predicts exceptional states again at $N' = 19, 23, 31, \ldots$, beyond the range KW tested; $N' = 19$ is rigorous for generic couplings.
4. *The window edge resonates.* In chord language the dressing cost is a time-integrated BPS→continuum two-point function. It decays as $e^{-E_0\tau}\tau^{-3/2}$ in central sectors, as $\tau^{-1/2}$ in resonant ones, and tends to a plateau, which is the obstruction, at the edge. The chord theory predicts a $K^{-1/2}$ tail for the resonant dressing costs. The exact data follow it over four decades, with Hill exponent $0.43 \to 0.48$ and an amplitude within 13–32% that approaches the chord value as N' grows, without fitted parameters. The exact uplift keeps the resonance in the dressing cost, but not in the decoder angles.

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1 Introduction

1.1 Where we stood

The uplift question of CV [1] asks how a BPS state of the N-mode one-flavor $\mathcal{N} = 2$ SYK model [3] continues to a BPS state of the $(N+1)$-mode model, inside the fortuity window of CCSY [4]. CV answer it cohomologically, through the long exact sequence (CV eq. 24). Its connecting map is $\delta[\alpha] = [C'\alpha]$, where C' collects the couplings involving the new mode. The 13 October report [9] (derivation D6) solved the *metric* version, which asks for the harmonic representatives. It reduced the BPS conditions to the coupled uplift equations $a = h + \varepsilon K' b$, $b = g - \varepsilon K a$ (its eq. (4.5), with K, K' in its eq. (4.4)), and solved them explicitly through an anti-Hermitian linear system (its eqs. (5.3)–(5.4)). Eliminating b gives the closed form

$$\Psi = a + b \wedge e, \qquad a = (1 + K'K)^{-1}(h + \varepsilon K' g), \qquad b = g - \varepsilon K a, \qquad K = Q_0^+ E, \quad K' = (Q_0^\dagger)^+ E^\dagger. \tag{1.1}$$

Here h and g are old BPS states in the two occupation slots of the new mode e, $E = \eta\wedge$ is exterior multiplication by the $(q-1)$-form η of couplings touching e (so $E = C'$ up to sign conventions),

and $\varepsilon = (-1)^F$. Everything else in that report follows from (1.1): walls, atoms, the dressing cost as a two-point function, the closed-form decoder, and $T_4 = T_2 - H$.

The derivation used half filling exactly once. The equation $Q_0 b = -\varepsilon E a$ can be solved only if Ea is exact. At half filling this is automatic, because Ea has degree $F + q - 1$, whose level lies outside the window, where there is no cohomology. In a general sector of the window the target degree can carry BPS states, and Ea can have a harmonic part. This report treats every sector.

1.2 What this report does

- Section 2 reviews the Hodge decomposition per degree of Kanazawa–Wettig [2], which is the dimensional input everywhere below, in our conventions.
- Section 3 does the level bookkeeping, defines the three regimes, and presents the jump map.
- Section 4 proves that (1.1) holds in every sector on the admissible sources.
- Section 5 computes the connecting maps exactly, including the exchange and the exceptional states.
- Section 6 analyses the dressing in the three regimes, in chord language and at finite N.
- Section 7 collects the decoder statistics across the window.
- Section 8 discusses what is established and what is open.

All statements are tagged [derived], [derived + verified], [numerical], [derived from the chord rules] or [interpretation]. Appendices A–B list the checks and the runs.

2 Hodge pieces per degree (Kanazawa–Wettig)

KW [2] classify the level statistics of $\mathcal{N} = 2$ SYK one fermion-number sector at a time, and one Hodge piece at a time. For $q = 3$ (their $\hat{q}$; their N_c is our N) they split each sector as

$$V_f = V_f^+ \oplus V_f^- \oplus V_f^z \qquad \text{(KW eqs. 6.8–6.10)}, \tag{2.1}$$

[literature] with V^+ annihilated by $\bar{Q}$ but not by Q, V^- the reverse, and V^z the zero modes. Their Q raises f by 3, as our $Q = \omega\wedge$ does. In our language:

- V^+ is the *coexact* forms, V^- the *exact* forms, and V^z the BPS (harmonic) forms.
- Q maps V_f^+ bijectively onto V_{f+3}^- (KW eq. 6.12): this is the supermultiplet structure.
- Under the premise that all zero modes lie in the fortuity window (KW eq. 6.15, CV eq. 12), exact sequences give all three dimensions at every degree (KW eqs. 6.16–6.21, Table 8, App. B).

Our rank computations reproduce KW's App. B rows exactly for $N = 5, 9, 12, 13$ [numerical], including the exceptional zero modes (1 at $N = 5, 13$; 3 at $N = 9$). One notational trap: CV's App. A uses the opposite labels, CV's V^- being coexact and CV's V^+ exact; we checked this against KW's $N = 5$ row.

Two further results of KW matter below.

- [literature] *One special multiplet.* Every $V_f^\pm$ is GUE except the multiplet $V^+_{(N-3)/2} \leftrightarrow V^-_{(N+3)/2}$. It runs from the coexact forms at level -3 to the exact forms at level $+3$, straddling the window. For odd N it is GOE ($N \equiv 3 \bmod 4$) or GSE ($N \equiv 1 \bmod 4$) (KW eq. 6.13; general

$\hat{q}$ in eq. 6.24). It is also the "CT-invariant multiplet $(-\hat{q}/2, \hat{q}/2)$" of Turiaci–Witten's SYK check [7], which they find to be $\beta = 1$ at $N = 11$ and explain the same way (their Fig. 4 and pp. 18–19, with $J = f - N/2$).

- [literature] *Exceptional zero modes.* For $N \equiv 1 \pmod 4$ the block containing the window edges is in class DIII. Its Kramers degeneracy on an odd dimension forces at least two zero modes (KW §6.2). The six at $N = 9$ (rather than two) are unexplained (KW footnote 12).

Section 6 shows that the special multiplet is exactly the one through which off-centre uplifts dress. Section 5 gives a third derivation of the exceptional states, after KW's Kramers argument and CV's eq. (15).

3 Where the uplift jumps from and where it lands

3.1 Bookkeeping

Fix the new sector: fermion number F of $N' = N + 1$ modes, at level $\ell' = 2F - N'$, with $|\ell'| \le q$. Write $\Psi = a + b \wedge e$ as before.

- The $n_e = 0$ slot $a \in \Lambda^F$ sits at old level $\ell' + 1$. In CV's walk language [1] (§4) this is the *down step*.
- The $n_e = 1$ slot $b \in \Lambda^{F-1}$ sits at old level $\ell' - 1$: the *up step*.
- E raises the level by $2(q-1)$, $E^\dagger$ lowers it by $2(q-1)$, and $Q_0^\pm$ shifts it by $\pm 2q$.

The first-order dressing of an $n_e = 0$ source h is $Kh = Q_0^+ Eh$. Its path is

$$\ell' + 1 \xrightarrow{E} \ell' + 2q - 1 \xrightarrow{Q_0^+} \ell' - 1, \tag{3.1}$$

so the cloud lands in the other slot. The supermultiplet traversed is coexact$(\ell'-1) \leftrightarrow$exact$(\ell'+2q-1)$. The mirror path for an $n_e = 1$ source g is $\ell' - 1 \to \ell' - 2q + 1 \to \ell' + 1$.

In the chord variables of the 14 October report [10], which follow BBNR [5], the charge is $s = \ell/2q$. The two dressing problems are BPS→continuum two-point functions at charges

$$h: \ (s_{\text{out}}, s_{\text{in}}) = \left(\tfrac{\ell'+1}{2q}, \tfrac{\ell'+2q-1}{2q}\right), \qquad g: \ (s_{\text{out}}, s_{\text{in}}) = \left(\tfrac{1-\ell'}{2q}, \tfrac{2q-1-\ell'}{2q}\right). \tag{3.2}$$

At $q = 3$ and half filling these are the 14 October values $(\frac{1}{6}, \frac{5}{6})$ for even N', and $(0, \frac{2}{3})$, $(\frac{1}{3}, 1)$ for odd N'.

Relation to CV's walks. CV App. A counts the ordinary BPS states at (N', ℓ') by walks confined to the strip $|\ell| \le q - 1$. Appending the last step gives the recursion

$$S_{N'}(\ell') = S_N(\ell' + 1) + S_N(\ell' - 1), \qquad S_N(\ell) = 0 \ \text{ for } |\ell| \ge q, \tag{3.3}$$

which is the dimension count of Theorem 1 below whenever $\delta = 0$ and no exceptional states are involved. Section 5 shows exactly where this walk-by-walk picture must be amended: at the edge (the radical) and one step inside it (the exchange).

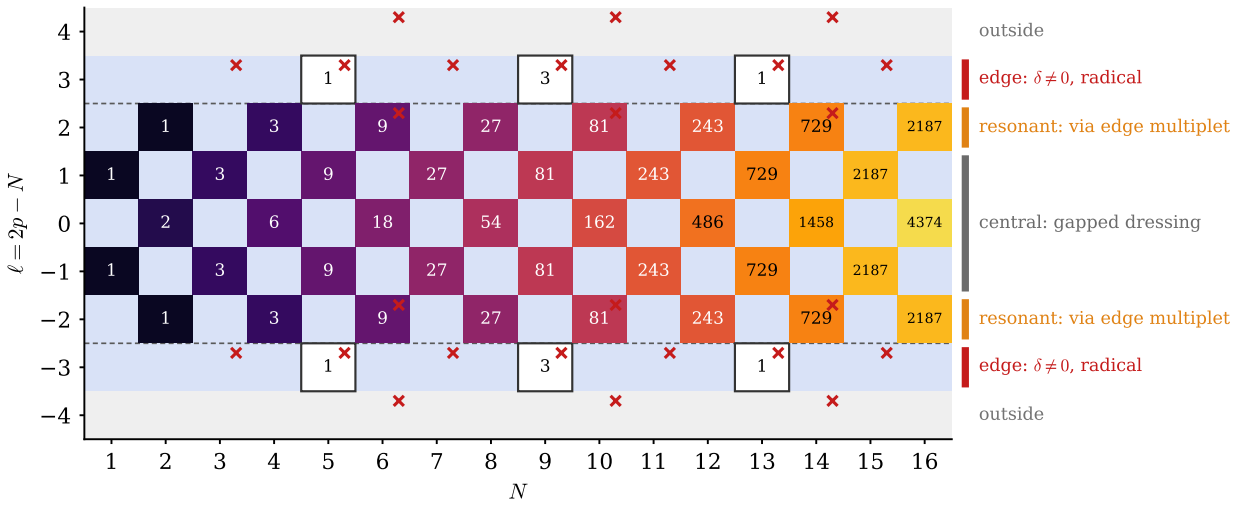


Figure 1: BPS dimensions of the $q = 3$ model in the $(N, \ell = 2p - N)$ plane, in the conventions of CV Fig. 2. Light blue: the fortuity window $|\ell| \leq 3$; dashed lines: the restricted window; white squares: exceptional states. Right margin: the regime of the uplift *into* each row ℓ' (Sec. 3). Red crosses: sectors whose connecting map δ is nonzero. These are the edge sectors $\ell' = \pm 3$ (always), $\ell' = \pm 2$ when $N = N' - 1 \equiv 1 \pmod 4$ (the exchange, Sec. 5), and $\ell' = \pm 4$, where the old exceptional states are obstructed from stepping outward. All dimensions were checked by direct computation for $N' \leq 14$ in every sector, and through the admissible-source count at $N' = 15, 16$.

3.2 Three regimes

Where the target level $\ell' \pm (2q - 1)$ sits relative to the window (equivalently, s_{in} against $\frac{1}{2}$) gives the classification [derived]:

regime	dressing target	s_{in}	h family at	g family at
gapped	outside the window	$> \frac{1}{2}$	$\ell' \geq 2 - q$	$\ell' \leq q - 2$
resonant	the window edge $\pm q$	$= \frac{1}{2}$	$\ell' = 1 - q$	$\ell' = q - 1$
obstructed	inside the window	$< \frac{1}{2}$	$\ell' = -q$	$\ell' = q$

This gives three kinds of sector.

- *Central* sectors $|\ell'| \leq q - 2$: both families are gapped. This is the 13 October regime.
- *Off-centre* sectors $|\ell'| = q - 1$: one family is resonant.
- *Edge* sectors $|\ell'| = q$: one family is obstructed, and the other slot is empty because it lies outside the window.

For $q = 3$: the central sectors are $\ell' \in \{0, \pm 1\}$, the resonant ones $\ell' = \pm 2$ (even N' only), and the edge ones $\ell' = \pm 3$ (odd N' only). This refines CV's necessary conditions (34)–(35) for a nonzero connecting map, $\ell \leq 2 - q$ or $\ell \geq q - 2$ on the old class, which say precisely that the target is inside the window or on its edge. Figure 1 draws the classification on CV's BPS plane, and Fig. 2 draws one uplift of each kind.

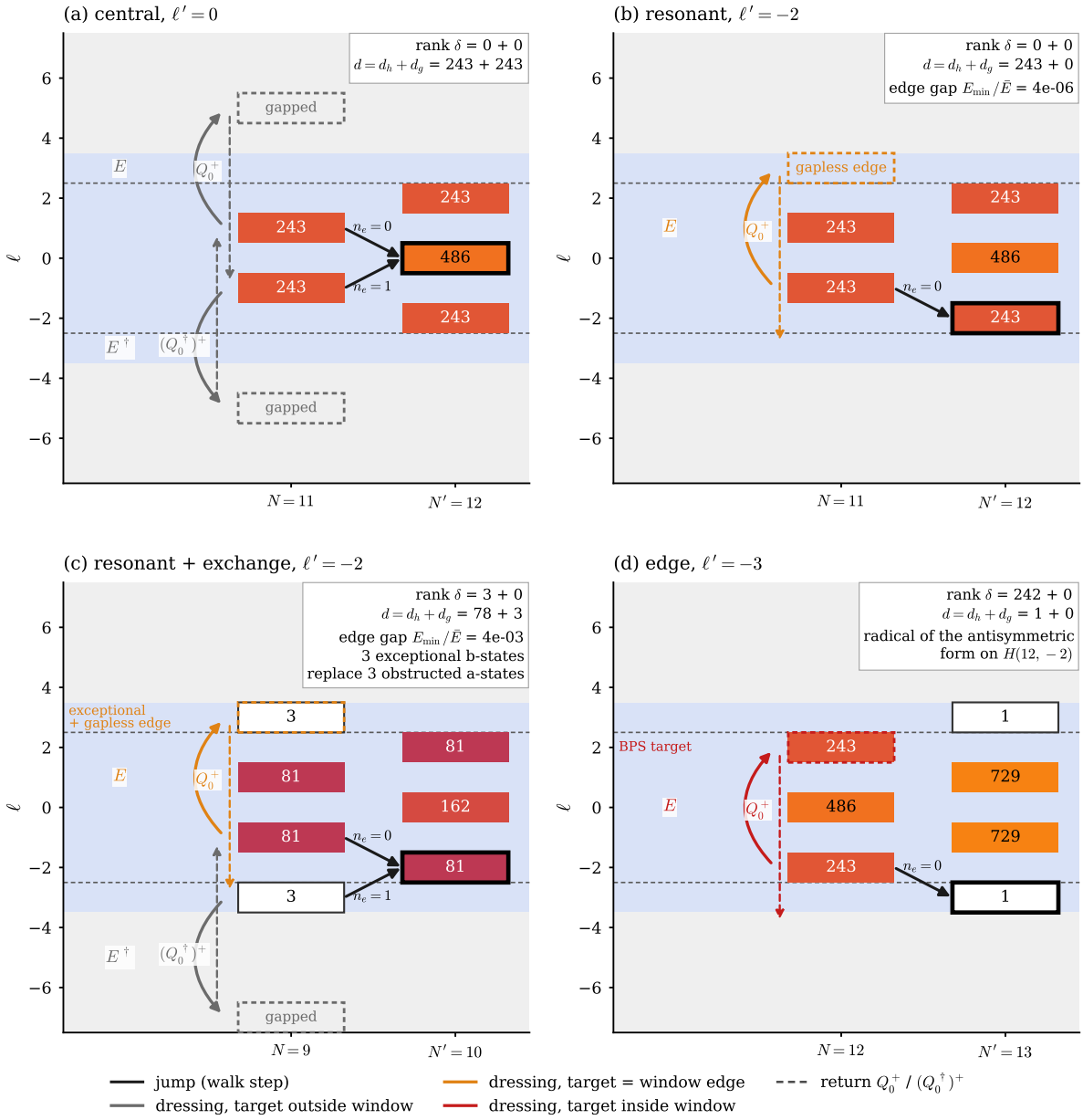


Figure 2: The four kinds of uplift, each drawn as a single step $N \to N'$ (columns), with the old and new BPS dimensions. Black arrows: the two jumps (walk steps) from the slots at $\ell' \pm 1$ into the target sector (bold frame). Solid coloured arcs: the first-order dressing by E (from the $n_e = 0$ slot) and $E^\dagger$ (from the $n_e = 1$ slot) to their target levels (dashed boxes). Dashed arrows: the return through Q_0^+ or $(Q_0^\dagger)^+$ into the other slot. Colour gives the regime of the target: grey outside the window, orange the window edge, red inside the window. The boxes give the measured rank of δ, the admissible dimensions $d_h + d_g$, and, for resonant sectors, the gap $E_{\min}/\bar{E}$ of the edge multiplet (seed 9000). (a) Central: both loops are gapped. (b) Resonant: the E loop passes through the gapless edge multiplet at $+3$. (c) Resonant with exchange ($N = 9$): δ maps three ordinary classes at -1 onto the three exceptional states at $+3$, and the three exceptional states at -3 take their place through the up step. (d) Edge: E lands on the BPS states at $+2$, and only the radical of an antisymmetric form survives, giving the exceptional state of $N' = 13$.

4 The exact uplift in every sector

Lemma 1 ([derived]). *For any q odd, E commutes with Q_0 and $E^\dagger$ with $Q_0^\dagger$, because η has even degree $q-1$. Hence:*

1. *E maps closed forms to closed forms and exact forms to exact forms ($EQ_0x = Q_0Ex$);*
2. *$E^\dagger$ maps coclosed forms to coclosed forms and coexact forms to coexact forms;*
3. *for a BPS state with $a = h+x$ (h harmonic, x exact) and $b = g+z$ (z coexact), $P_HEa = P_HEh$ and $P_HE^\dagger b = P_HE^\dagger g$.*

Theorem 1 ([derived + verified]). *Define the connecting maps $\delta_h = P_HEP_H$ on the harmonic forms H_X of the $n_e = 0$ slot and $\delta_g = P_HE^\dagger P_H$ on H_Y of the $n_e = 1$ slot, and the admissible sources $A_h = \ker\delta_h$, $A_g = \ker\delta_g$. Then*

$$B(N', F) = \Psi(A_h, 0) \oplus \Psi(0, A_g), \tag{4.1}$$

with Ψ given by (1.1), and the sum is orthogonal. The map $(h, g) \mapsto \Psi(h, g)$ is an isomorphism, and $\dim B = \dim\ker\delta_h + \dim\ker\delta_g$, which is CV eq. (26).

Proof.

- The BPS equations (13 October, §2) are $Q_0a = 0$, $Q_0^\dagger b = 0$, $Q_0b = -\varepsilon Ea$ and $Q_0^\dagger a = \varepsilon E^\dagger b$, independently of the sector.
- By Lemma 1, Ea is closed. Its harmonic part is P_HEh. So the third equation is solvable iff $h \in A_h$, and similarly the fourth iff $g \in A_g$.
- On admissible sources the solution is the anti-Hermitian system of 13 October §5. It has a unique solution in any sector, because K' maps into exact forms and $K'K|_{\mathrm{Ex}} = L^\dagger L \geq 0$.
- Injectivity: $h = P_Ha$ and $g = P_Hb$.
- Orthogonality: the 13 October proof uses only Lemma 1 and the Hodge decomposition. □

So every closed form of 13 October holds in every sector once $B_o^F \to A_h$ and $B_o^{F-1} \to A_g$:

- the three operators K_H, K'_H, L with $P_{\mathrm{Ex}}K'P_{\mathrm{Co}} = L^\dagger$;
- the orthogonal splitting and the minimum-norm uplift;
- the decoder as a generalized eigenproblem $\mathcal{N}v = \mu\mathcal{G}v$;
- monotone self-dressing;
- the identity $T_4 = T_2 - H$.

Proposition 1 ([derived + verified]: mirror symmetry). *For every realization,* $\operatorname{spec}\lambda(N', F) = 1 - \operatorname{spec}\lambda(N', N' - F)$.

The Hodge star maps the harmonic space of the couplings C at degree F onto that of $\bar{C}$ at degree $N' - F$, and it exchanges $n_e \leftrightarrow 1 - n_e$. Complex conjugation then maps the harmonic space of $\bar{C}$ back to that of C without changing n_e. The 13 October symmetry at even N' is the self-mirror case $\ell' = 0$.

Verification [numerical]. With seeds 9000–9002, all sectors $|\ell'| \le q+1$, $q=3$ at $N'=5$–14 and $q=5$ at $N'=6$–14:

- $d_h + d_g$ equals the BPS dimension computed directly in the N'-mode model, which equals the known count;
- the uplift residual outside that space is $\le 6.4 \times 10^{-13}$, with full rank;
- the closed-form decoder agrees with the direct decoder to $\le 2.6 \times 10^{-11}$;
- the splitting overlaps are $\le 1.1 \times 10^{-13}$;
- Proposition 1 holds on 39 ($q=3$) and 40 ($q=5$) sector pairs to $\le 2.7 \times 10^{-12}$.

At $N'=15$ the general code reproduces the exact half-filling T_2 of 13 October (0.5866 and 0.6013 for the two seeds). At $N'=16$ it gives $T_2 = 0.6075$ for seed 9000 at half filling, the exact value of the 13 October pre-registered test.

5 The connecting maps

5.1 Duality and exact ranks

Lemma 2 ([derived]). *For harmonic α at level ℓ and β at level $-\ell - 2(q-1)$ of the N-mode model, define the top-form pairing*

$$B_\eta(\beta, \alpha) = \int \beta \wedge \eta \wedge \alpha. \tag{5.1}$$

Then:

1. *$\beta \mapsto \overline{\star\beta}$ is an antilinear bijection between harmonic spaces at levels $\pm\ell$, so the matrix of δ on $H(N,\ell)$ is that of B_η;*
2. *$\operatorname{rank} \delta|_{H(\ell)} = \operatorname{rank} \delta|_{H(-\ell-2(q-1))}$;*
3. *on the self-paired level $\ell = 1-q$, with $k = (N-q+1)/2$, $B_\eta(\alpha,\beta) = (-1)^k B_\eta(\beta,\alpha)$.*

Part 3 holds because η has even degree, so moving β (degree k) to the front costs $(-1)^{k^2} = (-1)^k$.

Proposition 2 (exact ranks, $q=3$; [derived] given the window premise, [numerical] verified)**.** *Let $e(N)$ be the number of exceptional states of the N-mode model per edge. For the uplift into level ℓ':*

ℓ'	δ_h: *source* $\to$ *target*	*rank* δ_h	A_h	A_g
$0, \pm 1$	*target outside window*	*0*	H_X	H_Y
-2	$H(N,-1) \to H(N,+3)$	$e(N)$, *onto*	*codim* $e(N)$	$H(N,-3)$ ($e(N)$ *exceptional*)
-3	$H(N,-2) \to H(N,+2)$	$3^{N/2-1} - e(N')$	*radical of* B_η	*0*
-4	$H(N,-3) \to H(N,+1)$	$e(N)$, *injective*	*0*	*0*

Positive ℓ' is the mirror, with δ_g.

Derivation.

- Row $\ell' = -4$ restates the window premise at N': there are no BPS states at $|\ell'| > q$ (KW eq. 6.15; CV eq. 12).
- Lemma 2 pairs $H(N,-3)$ with $H(N,-1)$, so row $\ell' = -2$ has the same rank as row $\ell' = -4$. This is the surjectivity.
- Row $\ell' = -3$ is the self-paired form.

[numerical] All 142 $q=3$ sector records ($N' \le 16$) and all 114 $q=5$ records agree with these ranks. For $q=5$ the same table holds with levels $\pm 4, \pm 5, \pm 6$ in place of $\pm 2, \pm 3, \pm 4$.

The exchange [derived]. At $\ell' = -2$ with $N \equiv 1 \pmod 4$ there are two competing effects.

- $e(N)$ ordinary classes of $H(N, -1)$ are obstructed: their dressing would have to land on the exceptional states at $+3$.
- The $e(N)$ exceptional states at -3 cannot step down (row $\ell' = -4$). They step up instead.

The dimension $3^{(N-1)/2}$ is unchanged, as CV's strip-walk count requires. But the uplift is not a walk-by-walk bijection: an exceptional class is created at the edge (next subsection), and leaves by an up step that displaces one ordinary class (Fig. 2c).

5.2 Exceptional states are a radical

At the edge $\ell' = -q$ the $n_e = 1$ slot is empty, so $B(N', -q) = \ker \delta_h$ on $H(N, 1-q)$. This is the radical of B_η on the self-paired level. An antisymmetric form on an odd-dimensional space always has a nonzero radical, which gives the following.

Theorem 2 ([derived])**.** *The exceptional BPS states of the N'-mode model at level $-q$ form the radical of B_η on $H(N'-1, 1-q)$. If $k = (N'-q)/2$ is odd and $\dim H(N'-1, 1-q)$ is odd, there is at least one.*

Remarks on Theorem 2.

- For $q = 3$, $\dim H(N, -2) = 3^{N/2-1}$ is odd, so the condition is $N' \equiv 1 \pmod 4$. This is the uplift form of KW's Kramers argument (KW §6.2) and of CV eq. (15).
- The parity is the same as in CV's version. CV's antisymmetric M on $\Lambda^a(N')$ gives $\dim \text{harm} \equiv \tilde{D}_a(N')$. Our condition involves $S_{N'-1}(1-q) = \tilde{D}_a(N'-1) - \tilde{D}_{a-1}(N'-1)$, while $\tilde{D}_a(N') = \tilde{D}_a(N'-1) + \tilde{D}_{a-1}(N'-1)$, in the notation of CV App. A.
- [numerical] The forms are exactly (anti)symmetric: $|B^T - (-1)^k B|/|B| \le 4.3 \times 10^{-15}$.
  - $q = 3$: radicals $1, 3, 1$ at $N = 4, 8, 12$, and 0 at $N = 6, 10, 14$.
  - $q = 5$: radical 1 at $N = 6, 10$. At $N = 14$ the form is antisymmetric but $\dim = 1000$ is even, and the radical vanishes, matching KW's "no exceptional modes at $N_c = 15$".
  - The singular values of the antisymmetric forms come in equal pairs.

Beyond KW's range [derived]. Theorem 2 forces exceptional states, for $N' \le 45$, at:

q	N' with forced exceptional states
3	all $N' \equiv 1 \pmod 4$
5	7, 11, 19, 23, 31, 35, 43
7	9, 21, 25, 29, 37
9	11, 15, 23, 39, 43

KW tested $\hat{q} = 5$ only up to $N_c = 17$, finding edge modes only at 7 and 11. For $N' = 19$ the statement is rigorous for generic couplings:

- CV's M on $\Lambda^7(19)$ is antisymmetric on the even dimension 50388;
- Q is injective on $\Lambda^2(19)$: rank 171 for two realizations, with $\sigma_{\min}/\sigma_{\max} = 0.89$, which implies the same for generic couplings;
- so the exact 7-forms have dimension 171, and $\dim H^7 \equiv 0 - 171 \equiv 1 \pmod 2$.

Figure 3 shows the $q = 5$ plane with its regimes.

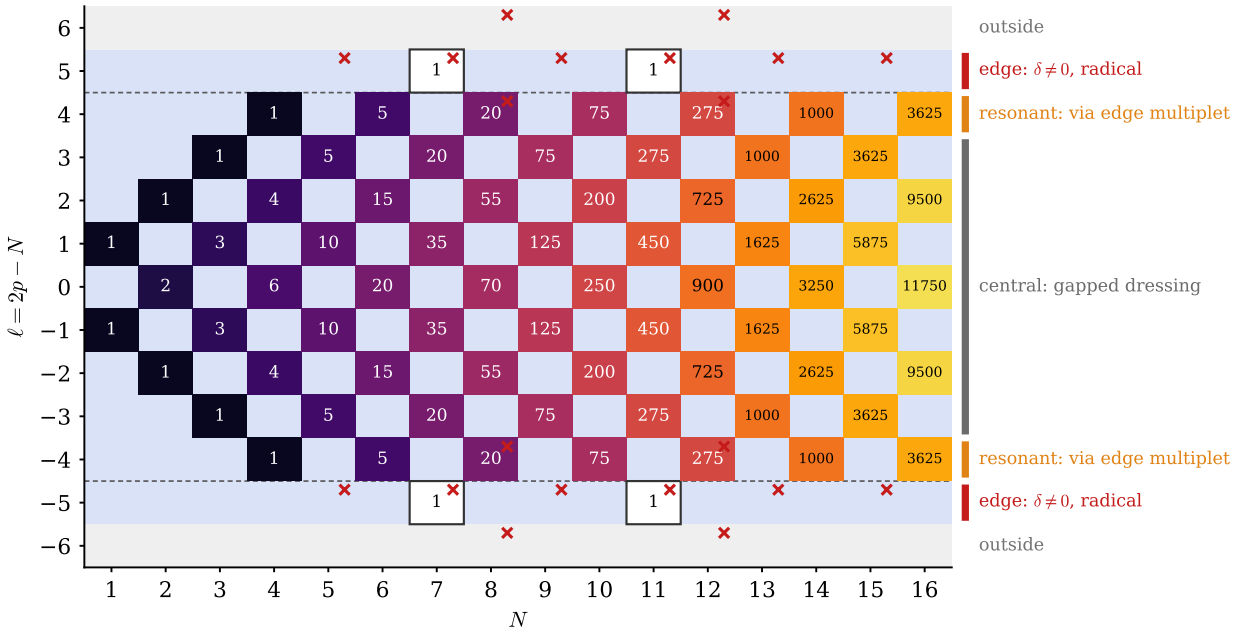


Figure 3: As Fig. 1 for $q = 5$, to be compared with CV Fig. 3. Every entry up to $N = 13$ agrees with CV. The resonant rows are $\ell' = \pm 4$ and the edge rows $\ell' = \pm 5$. Exceptional states appear at $N = 7, 11$ (KW). The parity criterion puts the next ones at $N = 19$.

Not explained. At $N' = 9$ ($q = 3$) the radical has dimension 3, not 1: this is KW's excess. The radical *subspace* moves with η (principal-angle cosines between different η are 0.05–0.49), but its dimension is 3 for every η we tried. We have no structural reason for this.

6 Dressing: gapped, resonant, obstructed

6.1 The two-point function in the three regimes

By the 13 October report, the first-order dressing cost of an admissible source is a BPS-projected, Euclidean-time-integrated two-point function of E:

$$\sigma^2 \in \operatorname{spec} PE^\dagger H_0^+ EP = \operatorname{spec} \int_0^\infty d\tau\, PE^\dagger e^{-\tau H_0} EP \qquad (\text{on } \ker \delta). \tag{6.1}$$

For a *non-admissible* source the integrand tends to $\|\delta h\|^2$ at late times, and the integral grows linearly. **The obstruction is the late-time plateau of the two-point function**, which in chord language is the BPS→BPS weight: the point mass of the 14 October Al-Salam–Chihara measure, equal to BLY eq. (4.14) [6].

In the chord limit, the 14 October closed form gives [derived from the chord rules]

$$G(\tau) = \int_0^\pi d\phi\, f(\phi)\, e^{-\tau\kappa\Lambda(\phi)}, \qquad \Lambda(\phi) = 2q^{-1/2}\big[\cosh\lambda(s_{\text{in}} - \tfrac{1}{2}) - \cos\phi\big], \tag{6.2}$$

$$f(\phi) = \frac{(ab;q^2)_\infty (q^2, e^{\pm 2i\phi}; q^2)_\infty}{2\pi (ae^{\pm i\phi}, be^{\pm i\phi}; q^2)_\infty}, \tag{6.3}$$

with $a = q^{s_{\text{in}} - 1/2}$ and $b = q^{1/2+\delta-s_{\text{out}}}$. Its late-time behaviour depends on the regime.

- $s_{\text{in}} > \frac{1}{2}$ (gapped): $a < 1$, so $f \propto \phi^2$ near $\phi = 0$, while $\Lambda \simeq \Lambda_{\min} + q^{-1/2}\phi^2$. Hence $G(\tau) \sim \tau^{-3/2} e^{-\kappa \Lambda_{\min} \tau}$.
- $s_{\text{in}} = \frac{1}{2}$ (resonant): $a = 1$. The zero of $(e^{\pm 2i\phi}; q^2)$ is cancelled by that of $(e^{\pm i\phi}; q^2)$, so $f(0) = 2(q^2; q^2)_\infty / [\pi (b; q^2)_\infty]$ is finite, while $\Lambda \simeq q^{-1/2}\phi^2$. Hence

$$G(\tau) \simeq \frac{f(0)}{2} \sqrt{\frac{\pi q^{1/2}}{\kappa \tau}} \;\propto\; \tau^{-1/2}, \tag{6.4}$$

  and $\int G \, d\tau$ diverges. Consistently, the 14 October closed form $\propto {}_2\phi_1(q^2, q^2; q^{2+2\delta}; q^2, q^{2s_{\text{in}}-1})$ has terms that tend to a constant as $s_{\text{in}} \to \frac{1}{2}^+$, which gives a pole $\propto 1/\lambda(2s_{\text{in}} - 1)$.
- $s_{\text{in}} < \frac{1}{2}$ (obstructed): a point mass at zero energy, i.e. the plateau.

6.2 The resonant tail

The 14 October report showed that, for uncrossed chord diagrams, the first-order dressing operator is free compound Poisson. Its jump law is $g(\phi) \propto A(\phi)/\Lambda(\phi)$ under the B-family density, at rate r = (inner modes)/(sources). At $s_{\text{in}} = \frac{1}{2}$ the jumps are unbounded: $g \simeq G_0/\phi^2$ as $\phi \to 0$, while the density tends to a finite ρ_0. A single large jump produces a single large eigenvalue, so [derived from the chord rules]

$$P(\sigma^2 > K) \simeq r \, w(g > K) \simeq r \rho_0 \sqrt{G_0} \, K^{-1/2}. \tag{6.5}$$

The mean is therefore infinite in the chord limit; at finite N the smallest inner energy cuts it off. The amplitude is parameter-free. From the 14 October chord functions at $(s_{\text{out}}, s_{\text{in}}) = (-\frac{1}{6}, \frac{1}{2})$: $P\sqrt{K} = 0.50, 0.51, 0.52$ at $N' = 12, 14, 16$.

6.3 Finite N

- [derived] *Which multiplet.* By the bookkeeping (3.1), the resonant family dresses through coexact$(-q)$ $\leftrightarrow$exact$(+q)$, which is exactly KW's GOE/GSE multiplet. [interpretation] It is also the only multiplet in which Q_0 maps between ambient spaces of equal dimension, $\binom{N}{(N-q)/2} = \binom{N}{(N+q)/2}$. That would explain a "square" hard edge at zero.
- [numerical] *Its gap.* The seed means are $E_{\min}/\bar{E} = 2.5 \times 10^{-1}$, 2.1×10^{-2}, 2.9×10^{-3}, 5.9×10^{-6}, 1.6×10^{-6} at $N = 5, 7, 9, 11, 13$, and 8.0×10^{-8} and 1.5×10^{-10} (two seeds) at $N = 15$ (Fig. 4a). Gapped families stay at 0.02–0.6. In units of the multiplet dimension n $(= 4, 21, 80, 319, 1208, 4551)$, $n^2 E_{\min}/\bar{E}$ stays at 0.003–24 per seed (0.15–13 for $q = 5$). This is the hard-edge scaling of a square random matrix [interpretation]: $E_{\min}$ is exponentially small in N because n is exponentially large. The smallest values (0.003 and 0.06) occur in the GOE multiplets ($N \equiv 3 \bmod 4$), as the softer $\beta = 1$ hard edge ($P(n\sigma_{\min} < x) \propto x$) would suggest; the Kramers (GSE) multiplets never go below 1.1. With 2–3 seeds per N this is suggestive only.
- [numerical] *Dominance of the bottom.* The lowest 1% of inner modes carries 90%, 96% and 99% of the first-order dressing trace at $N = 11, 13, 15$, against $\leq 14\%$ in gapped families.
- [numerical] *Dressing costs.* The seed-mean is 1.3, 4.1, 7.0, 1244, 851, 1579 and 6.6×10^5 (two seeds; the latter from one mode with $E/\bar{E} = 1.5 \times 10^{-10}$, and 613 and 688 without the top state) at $N' = 6, 8, 10, 12, 14, 16$; the median 0.14–0.74; the maximum 1.45×10^9. Central sectors have mean ≤ 0.42 and maximum ≤ 5.2. Single modes dominate the mean, which therefore fluctuates strongly from seed to seed, as a $K^{-1/2}$ tail requires.

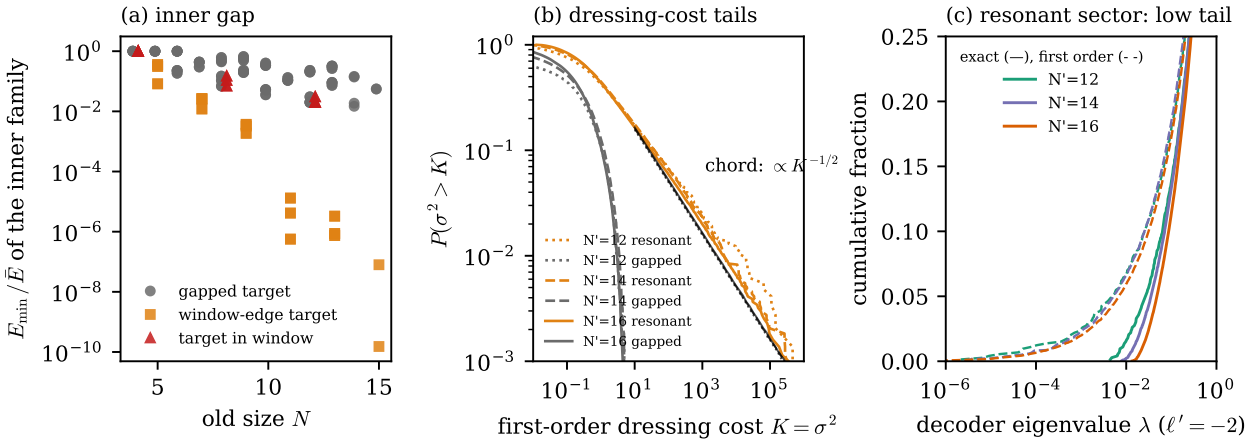


Figure 4: The window-edge resonance. (a) Gap $E_{\min}/\bar{E}$ of the inner family (the old energies through which each dressing family passes) against the old size N, coloured by regime. The window-edge multiplet (orange) closes to $\sim 10^{-6}$; gapped families do not. (b) Fraction of sources with first-order dressing cost above K. Orange: resonant families ($\ell' = \pm 2$); grey: central families. Black: the parameter-free chord tail $r\,w(g > K)$ of Eq. (6.5) at the same N', $\propto K^{-1/2}$. (c) Cumulative distribution of the lowest decoder eigenvalues in the resonant sector $\ell' = -2$: first order (dashed) against exact (solid). First order places resonant states down to $\lambda \sim 10^{-7}$–10^{-9} (clipped at 10^{-7}); the exact decoder stops near 10^{-2}. All panels pool the available seeds (3; 2 at $N' = 16$).

- [numerical] *The tail* (Fig. 4b). $P(\sigma^2 > K) \propto K^{-\alpha}$ over four decades.
  - The Hill estimate of α (for $K > 100$; h family at $\ell' = -2$, all seeds) is 0.43 ± 0.06 at $N' = 12$ ($n = 48$), 0.48 ± 0.04 at $N' = 14$ ($n = 144$), and 0.48 ± 0.03 ($n = 259$) at $N' = 16$.
  - The chord value (6.5) is $\alpha = \frac{1}{2}$. The measured amplitude at $K = 100$ is $P\sqrt{K} = 0.66$, 0.66, 0.59 at $N' = 12, 14, 16$, against the chord 0.50, 0.51, 0.52.
  - The ratio of data to chord amplitude falls with N' (1.32, 1.29, 1.13), toward the chord law.
- [numerical] $q = 5$. The same happens at $|\ell'| = 4$ and nowhere else: gap 8.1×10^{-5} ($N = 11$) and 7.5×10^{-7} ($N = 13$); mean dressing 98 and 1070.

6.4 What the exact uplift does with the resonance

Write $Q_0 = \sum_j s_j |u_j\rangle\langle v_j|$ on the inner multiplet, and $e = U^\dagger E V_h$. The self-dressed metric block of 13 October is then exactly [derived]

$$K_H^\dagger (1+R)^{-1} K_H = e^\dagger (S^2 + W)^{-1} e, \qquad W = U^\dagger E P_{\mathrm{Ex}} E^\dagger U = SRS. \tag{6.6}$$

This is the first-order dressing with the propagator of $H_0 + E P_{\mathrm{Ex}} E^\dagger$ in place of H_0^{-1}: a Dyson resummation whose self-energy is $E P_{\mathrm{Ex}} E^\dagger$ (verified to 4×10^{-15}).

But $[E, Q_0] = 0$ forces the self-energy to be "soft": $W_{jj} = s_j^2 R_{jj}$, with R_{jj} bounded (measured 0.06–0.14 on the lowest modes). It vanishes on the near-zero modes. **So the resonance survives in the exact minimum-norm dressing cost** [numerical]: the mean is 1190 against 1244 at first order ($N' = 12$), 787 against 851 ($N' = 14$), and 1454 against 1579 and 6.1×10^5 against 6.6×10^5 (two seeds) ($N' = 16$).

The decoder angles do not resonate. For a state whose cloud dominates, the exact h block gives [derived]

$$\lambda \to 1 - \frac{\langle \hat{k}, (1+R)^{-2}\hat{k}\rangle}{\langle \hat{k}, (1+R)^{-1}\hat{k}\rangle}, \tag{6.7}$$

which is set by the bounded operator R, not by σ. [numerical] First order puts 2.0–2.5% of resonant-sector states within 10^{-3} of the wall, its lowest eigenvalue being 10^{-7}–10^{-5}. The exact decoder has none there, and at $N' = 16$ none within 10^{-2}. Its floor rises with N': the lowest eigenvalue is $(4.5\text{–}6.5) \times 10^{-3}$ at $N' = 12$, $(8\text{–}10) \times 10^{-3}$ at $N' = 14$, and $(1.4\text{–}1.5) \times 10^{-2}$ (first order: 5×10^{-7}, 7×10^{-10}) at $N' = 16$ (Fig. 4c). The 13 October per-mode mean field, which replaces R by its diagonal, fails here: its T_2 is low by 0.04–0.05. The spread of R along the resonant direction matters.

Reading [interpretation].

- In resonant sectors, a few BPS states of N' are made mostly of the near-zero modes of the old edge multiplet, i.e. of states that are almost BPS at the window edge $s = \pm\frac{1}{2}$.
- Their harmonic seed carries a vanishing fraction of the norm, $\|h\|^2/\|\Psi\|^2 \sim 1/\kappa$. This is CV's metric fragility in its strongest form, localized at $|\ell'| = q - 1$.
- In chord language, these sectors are where the uplift's matter chord ends on the gapless $Q = \frac{1}{2}$ supermultiplet of the super-Schwarzian. This is HITZ App. A [8], as already used for two flavors in the 25 October report [11], whose resonances at $s' = \pm\frac{1}{5}$ are the analogue.

7 The decoder across the window

N'	ℓ'	d (h+g)	T_2: first → self → exact	H: first / exact	$T_4 - (T_2 - H_{\text{first}})$	walls(10^{-3}): first / exact
12	0	486 (243+243)	0.631 → 0.670 → 0.685	0.085 / 0.090	−0.0045	0.329 / 0.329
12	−2	243 (243+0)	0.515 → 0.497 → 0.497	0.125 / 0.132	−0.0069	0.025 / 0.000
13	−1	729 (486+243)	0.563 → 0.604 → 0.616	0.111 / 0.109	+0.0026	0.112 / 0.123
13	−3	1 (1+0)	$\lambda = 0.65, 0.58, 0.65$ (one exceptional state)	–	–	–
14	0	1458 (729+729)	0.566 → 0.619 → 0.636	0.105 / 0.108	−0.0034	0.116 / 0.128
14	−2	729 (728+1)	0.477 → 0.455 → 0.455	0.133 / 0.142	−0.0092	0.023 / 0.000
16	0	4374 (2187+2187)	0.539 → 0.595 → 0.608 (1 seed)	0.122 / 0.125	−0.0030	0.011 / 0.021
16	−2	2187 (2187+0)	0.452 → 0.427 → 0.427	0.138 / 0.148	−0.0106	0.020 / 0.000

Table 1: Decoder statistics by sector (3 seeds; at $N' = 16$, 2 seeds). "Self" is the exact uplift without the h–g mixing. Walls: the fraction with $\min(\lambda, 1-\lambda) < 10^{-3}$. In central sectors these are the exact atoms of 13 October; in resonant sectors they are first-order artifacts.

- [numerical] *The T_4 prediction.* $T_4 = T_2 - H$ is exact. The first-order prediction $T_4 \simeq T_2 - H_{\text{first}}$ of 13 October holds to ≤ 0.009 in every extensive sector at $N' = 9$–14, and to ≤ 0.011 at $N' = 16$ (Fig. 5b, Table 1). It is not uniform across the window, though.
  - In resonant sectors the error grows steadily: -0.0069, -0.0092, -0.0106 at $N' = 12, 14, 16$.
  - In central sectors it shrinks: -0.0045, -0.0034, -0.0030.
  - The first-order spectrum of resonant sectors is visibly wrong at the bottom, so this degradation is expected.
- [numerical] *Self-dressing lowers T_2 in resonant sectors.* Cloud-dominated states move away from the $\lambda = 0$ wall, so $y = \mu^2$ drops from about 1. In central sectors self-dressing raises T_2.

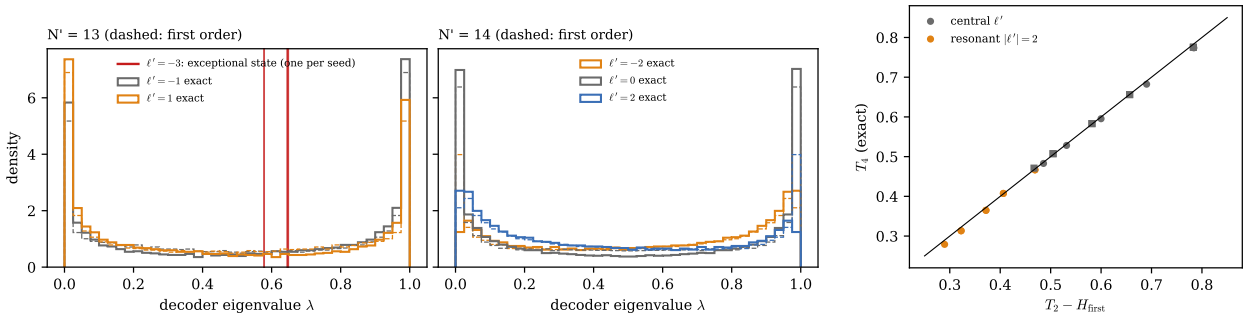


Figure 5: (a) Decoder densities at $N' = 13$ (left) and $N' = 14$ (middle) by sector; solid exact, dashed first order. Mirror sectors are reflections of each other. At $N' = 13$ the vertical lines are the single exceptional state at $\ell' = -3$ (one per seed). (b) Right: exact T_4 against the first-order prediction $T_2 - H_{\text{first}}$. Circles are even N', squares odd N', for all extensive sectors with $N' = 6$–16; the line is $T_4 = T_2 - H_{\text{first}}$.

- [derived]+[numerical] *Atoms.* The 13 October count generalizes: forced atoms $= \max(0, \dim A - \dim(\text{target family}))$. At $\ell' = -2$ the target family is the edge multiplet. At $N' = 8$ this forces $27 - 21 = 6$ atoms, observed in every seed; none are forced from $N' = 10$ on.
- [numerical] *Exceptional states.* At $\ell' = -q$ they are h-type, carrying 58–76% of their weight in $n_e = 0$ at $N' = 5, 13$ (0.27–0.79 at $N' = 9$).
- [numerical] *The exceptional states that uplift at* $\ell' = -2$ (the exchange) have gapped dressing, with $\lambda_1 = 0.02$–0.13. At $N' = 10$ they remain separate states near $\lambda = 0.003$–0.02. At $N' = 14$ the single one dissolves into the resonant h block: no exact eigenvector has more than 5–9% weight on it.

8 Discussion

8.1 What changes off half filling

Three things change, and each has a precise location on the jump map.

1. *Cohomology* (CV's δ) acts only at the edge rows $|\ell'| = q$, at the exchange rows $|\ell'| = q - 1$ when the old model has exceptional states, and at $|\ell'| = q + 1$.
2. *Metric resonance* acts only at $|\ell'| = q - 1$.
3. Everywhere else the 13 October picture holds unchanged.

Exceptional states acquire a life cycle. They are born as a radical at the edge, and leave by displacing one ordinary class one step inside.

8.2 What is established, and how firmly

- *Exact.*
  - Theorem 1 and its corollaries, and Proposition 1;
  - the duality and (anti)symmetry of B_η, Theorem 2, and the parity criterion;
  - $q = 5$, $N' = 19$, given one rank computation;
  - the Dyson identity with $W = SRS$, and the cloud-dominated limit of λ.

- *Exact given the window premise* (KW 6.15, CV Conj. A.1, both verified to $N \leq 17$): the rank table and the exchange.
- *Derived from the chord rules:* the three late-time behaviours, and the $K^{-1/2}$ tail with its amplitude.
- *Numerical:*
  - the machine-precision verification;
  - the hard edge of the edge multiplet;
  - the tail exponent and amplitude;
  - the absence of a wall pile-up in the exact decoder;
  - $T_4 \simeq T_2 - H_{\text{first}}$ in every sector, though slowly degrading in resonant sectors.
- *Heuristic or interpretive:* the "square" explanation of the hard edge, and the reading of cloud-dominated states as BPS states assembled from the old edge continuum.

8.3 Limitations

- Sizes: $N' \leq 16$ with dense linear algebra, and 1–3 seeds per sector (2 for the $N' = 16$ resonant sector, 1 for the $N' = 16$ central sector).
- The resonant tail is measured over at most four decades, with $\lesssim 260$ events above $K = 100$. The amplitude agrees to 13–32%, improving with N', and the effective exponent is still drifting. A single near-zero mode can dominate the mean dressing cost of a realization (seed 9001 at $N' = 16$).
- The exponentially small edge gap makes numerical rank decisions delicate. At $N' = 16$ one record (seed 9001) first failed the dimension check: a coexact edge mode with $\sigma/\sigma_{\max} \sim 10^{-5}$ fell below a Gram-eigenvalue threshold. Ranks are now decided by singular values, consistently with the pseudo-inverse, and the record was rerun. Sizes beyond $N' \approx 18$ will need rank decisions informed by the known dimensions, or higher precision.
- The chord prediction (6.5) keeps only uncrossed diagrams and the rank-one large-jump approximation. The 14 October report measured crossed diagrams at $\leq 0.5\%$ of the third moment in central sectors. This has not been checked for heavy tails.

8.4 Open problems

- The rank deficit at $N' = 9$ (radical 3 instead of 1).
- The large-N laws of $E_{\min}$ and of the tail amplitude. Sparse or partial methods could reach $N' = 18$, since only the lowest edge modes matter.
- A closed form for the exact resonant decoder edge, $\lambda_{\min} \sim 10^{-2}$: the spectral measure of R along the resonant modes.
- The two-flavor counterpart of the radical criterion and of the exchange.
- Whether the cloud-dominated states have a direct gravitational description, as BPS states of N' built from the gapless edge continuum of N.

A Verification summary

check	result
KW App. B rows from our rank computations ($N = 5, 9, 12, 13$)	exact match
dimensions: admissible = direct = strip walks + exceptional ($q = 3$: $N' \leq 14$ direct, 15–16 admissible; $q = 5$: ≤ 14)	all sectors, all seeds
connecting-map ranks vs Proposition 2	142 ($q = 3$) + 114 ($q = 5$) records, all equal
uplift residual / rank	$\leq 6.4 \times 10^{-13}$ / full
closed-form vs direct decoder	$\leq 2.6 \times 10^{-11}$
orthogonal splitting	$\leq 1.1 \times 10^{-13}$
mirror symmetry (Prop. 1)	$\leq 2.7 \times 10^{-12}$ (79 pairs)
(anti)symmetry of B_η; radicals	$\leq 4.3 \times 10^{-15}$; $1, 3, 1$ / $1, 1, 0$ as predicted
$Q\|_{\Lambda^2(19)}$ injective ($q = 5$)	rank 171, $\sigma_{\min}/\sigma_{\max} = 0.89$
Dyson identity $W = SRS$	$\leq 4 \times 10^{-15}$
monotone self-dressing (single-family sectors)	50 sectors; violations $\leq 3 \times 10^{-11}$ (round-off)
T_2 at half filling vs 13 October	$N' = 15$: 0.5866, 0.6013 (D6: 0.587, 0.601); $N' = 16$: 0.6075 (= D6)
unit tests	`test_uplift_window.py` (9), `test_uplift.py` (6), `test_chord_dressing.py` (8), all in `tests/`

B Numerical methods and reproducibility

Code. All code is at commit `1ad5bb9`.

- `src/uplift.py`: `OldModel(F=)`, `harmonic_basis`, `uplift_bookkeeping`, `connecting_form`, `OldModel.dyson`. Half-filling behaviour is unchanged, and the existing tests pass.
- `src/chord_dressing.py`: `chord_sectors(F=)`.
- Scripts: `scripts/run_uplift_window.py` (tasks `sectors`, `forms`), `scripts/analyze_uplift_window.py` and `scripts/plot_uplift_window_map.py`.

Couplings. $c =$`random_couplings(default_rng(seed), C(N',q))`, shared by all sectors of one (N', seed). At $q = 3$ and half filling these are the realizations of the 13 October data (seeds 9000, 9001).

Runs. All under `scripts/memwatch.py`, with outputs in `results/data/uplift_window_2026-10-26/`:

- $q = 3$, $N' = 5$–14, seeds 9000–9002: peak 0.83 GB, 919 s;
- $q = 5$, $N' = 6$–14, seeds 9000–9001: peak 1.51 GB, 1048 s;
- connecting forms: peak 0.50 GB;
- $q = 3$, $N' = 15, 16$, $\ell' \leq 0$, seeds 9000–9001: the run that also did $N' = 15$ peaked at 7.82 GB (3653 s), and was stopped by hand after its last needed record; the seed-9001 rerun peaked at 5.42 GB (1261 s).

Two first launches crashed before writing any record (an empty-family bug, since fixed). One $N' = 16$ record was rerun after the rank-decision fix described in Sec. 8. Both incidents are recorded in `research/notes/uplift_window_2026-10-26.md`.

Analysis. `analyze_uplift_window.py` writes `summary.json` and `chord_tail.json`. The chord tail uses the 14 October functions at $(s_{\rm out}, s_{\rm in}, \delta) = (-\frac{1}{6}, \frac{1}{2}, \frac{2}{3})$ with 4000 Gauss–Legendre nodes.

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